Introduction to Computer Science

Lecture 1

Introduction to Computer

Lecture 1

OBJECTIVES AND INTENDED OUTCOMES

- By the end of this lesson the learner is expected to:
  - Understand the concept of computer, data, and information.
  - Learn the characteristics of computers.
  - Learn the computers classification.
  - Learn the computer application
  - Learn the Computer Generations

Computer Definition

- A computer is an electronic device through which data can be stored, processed, and then retrieved again when requested as information.
- A computer consists of two main components:
  - Hardware.
  - Software.

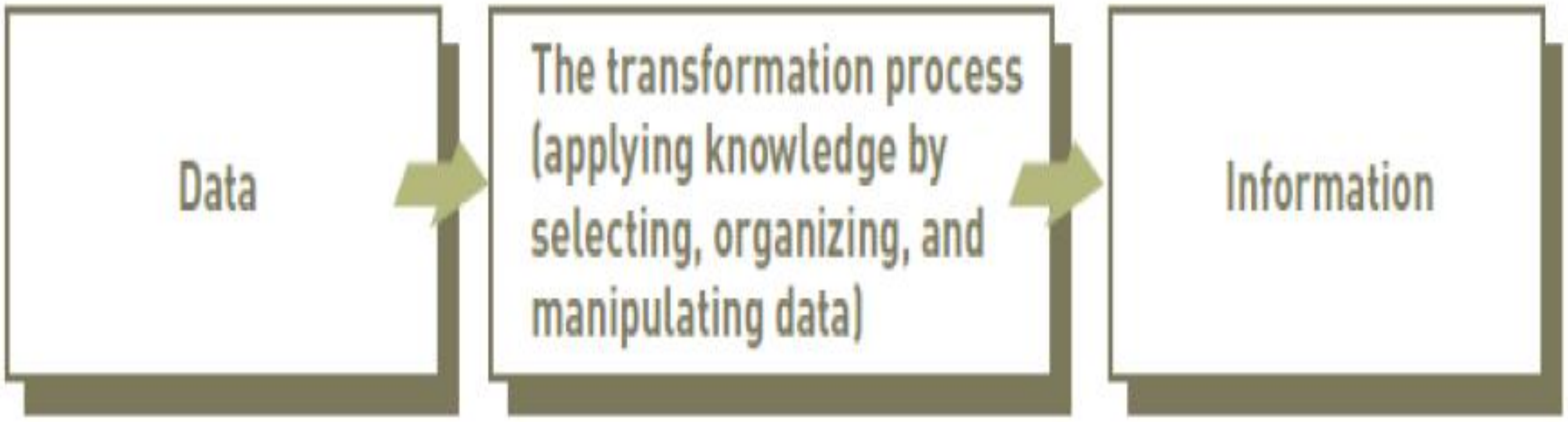
Data, Information, and Knowledge

- **Data** is the facts fed to the computer in order to process it to benefit from it by the user.
- **Information** is data that has been processed and analyzed to provide knowledge to decision makers and help them achieve certain goals.
- **Processing** is performing some manipulations on the entered data to give the desired results by the program.
  - storage-arrangement-classification-search-solving mathematical problems-graphics.
- **Knowledge** is Awareness and understanding of a set of information

Types of Data

Data	Represented by
Alphanumeric data	Numbers, letters, and other characters
Image data	Graphic images and pictures
Audio data	Sound, noise, or tones
Video data	Moving images or pictures

The Process of Transforming data into Information



Computer Characteristics

- Computers differ in size and price and agree on the main characteristics:
  - Super speed in carrying out operations.
  - Precision.
  - Storage possibility.
  - Multi-used

Classification of Computers

Type of Work

- Single purpose
- Multi-purpose

Working Method

- Analog
- Digital

Memory Size

Super Computers
Mainframe Computers
Mini Computers
Micro Computers

Computer Applications

- Computer use has become a measure of progress and development, as it is used in all fields.
  - official departments and institutions.
  - Trade and financial affairs
  - Universities and educational institutions
  - Airlines Civil
  - architectural engineering as building design services
  - The field of health in the management of hospitals

Computer Generations

Lecture1 Part2

Computer Generations

- The modern computer took its shape with the arrival of your time. It had been around 16th century when the evolution of the computer started. The initial computer faced many changes, obviously for the betterment. It continuously improved itself in terms of speed, accuracy, size, and price to urge the form of the fashionable day computer. This long period is often conveniently divided into the subsequent phases called computer generations:

Computer Generations

- First Generation Computers (1940-1956)
- Second Generation Computers (1956-1963)
- Third Generation Computers (1964-1971)
- Fourth Generation Computers (1971-Present)
- Fifth Generation Computers (Present and Beyond)

First Generation Computers: Vacuum Tubes (1940-1956)

- The technology behind the primary generation computers was a fragile glass device, which was called vacuum tubes. These computers were very heavy and really large in size. These weren’t very reliable and programming on them was a really tedious task as they used low-level programming language and used no OS. First-generation computers were used for calculation, storage, and control purpose. They were too bulky and large that they needed a full room and consume rot of electricity.

Main characteristics of first generation

Main electronic component	Vacuum tube.
Programming language	Machine language.
Main memory	Magnetic tapes and magnetic drums.
Input/output devices	Paper tape and punched cards.
Speed and size	Very slow and very large in size (often taking up entire room).
Examples of the first generation	IBM 650, IBM 701, ENIAC, UNIVAC1, etc.

Second Generation Computers: Transistors (1956-1963)

- Second-generation computers used the technology of transistors rather than bulky vacuum tubes. Another feature was the core storage. A transistor may be a device composed of semiconductor material that amplifies a sign or opens or closes a circuit.
- Transistors were invented in Bell Labs. The use of transistors made it possible to perform powerfully and with due speed. It reduced the dimensions and price and thankfully the warmth too, which was generated by vacuum tubes. Central Processing Unit (CPU), memory, programming language and input, and output units also came into the force within the second generation.
- Programming language was shifted from high level to programming language and made programming comparatively a simple task for programmers. Languages used for programming during this era were FORTRAN (1956), ALGOL (1958), and COBOL (1959).

Main characteristics of second generation

Main electronic component	Transistor.
Programming language	Machine language and assembly language.
Memory	Magnetic core and magnetic tape/disk.
Input/output devices	Magnetic tape and punched cards.
Power and size	Smaller in size, low power consumption, and generated less heat (in comparison with the first generation computers).
Examples of second generation	PDP-8, IBM1400 series, IBM 7090 and 7094, UNIVAC 1107, CDC 3600 etc.

Third Generation Computers: Integrated Circuits. (1964-1971)

- During the third generation, technology envisaged a shift from huge transistors to integrated circuits, also referred to as IC. Here a variety of transistors were placed on silicon chips, called semiconductors. The most feature of this era's computer was the speed and reliability. IC was made from silicon and also called silicon chips.
- A single IC, has many transistors, registers, and capacitors built on one thin slice of silicon. The value size was reduced and memory space and dealing efficiency were increased during this generation. Programming was now wiped out Higher level languages like BASIC (Beginners All-purpose Symbolic Instruction Code). Minicomputers find their shape during this era.

Main characteristics of third generation

Main electronic component	Integrated circuits (ICs)
Programming language	High-level language
Memory	Large magnetic core, magnetic tape/disk
Input / output devices	Magnetic tape, monitor, keyboard, printer, etc.
Examples of third generation	IBM 360, IBM 370, PDP-11, NCR 395, B6500, UNIVAC 1108, etc.

Fourth Generation Computers: Micro-processors (1971-Present)

- In 1971 First microprocessors were used, the large scale of integration LSI circuits built on one chip called microprocessors. The most advantage of this technology is that one microprocessor can contain all the circuits required to perform arithmetic, logic, and control functions on one chip.
- The computers using microchips were called microcomputers. This generation provided the even smaller size of computers, with larger capacities. That's not enough, then Very Large Scale Integrated (VLSI) circuits replaced LSI circuits. The Intel 4004chip, developed in 1971, located all the components of the pc from the central processing unit and memory to input/ output controls on one chip and allowed the dimensions to reduce drastically.
- Technologies like multiprocessing, multiprogramming, time-sharing, operating speed, and virtual memory made it a more user-friendly and customary device. The concept of private computers and computer networks came into being within the fourth generation.

Main characteristics of fourth generation

Main electronic component	Very large-scale integration (VLSI) and the microprocessor (VLSI has thousands of transistors on a single microchip).
Memory	semiconductor memory (such as RAM, ROM, etc.)
Input/output devices	pointing devices, optical scanning, keyboard, monitor, printer, etc.
Examples of fourth generation	IBM PC, STAR 1000, APPLE II, Apple Macintosh, Alter 8800, etc.

Fifth Generation Computers

- The technology behind the fifth generation of computers is AI. It allows computers to behave like humans. It is often seen in programs like voice recognition, area of medicines, and entertainment. Within the field of games playing also it's shown remarkable performance where computers are capable of beating human competitors.
- The speed is highest, size is that the smallest and area of use has remarkably increased within the fifth generation computers. Though not a hundred percent AI has been achieved to date but keeping in sight the present developments, it is often said that this dream also will become a reality very soon.
- In order to summarize the features of varied generations of computers, it is often said that a big improvement has been seen as far because the speed and accuracy of functioning care, but if we mention the dimensions, it's being small over the years. The value is additionally diminishing and reliability is in fact increasing.

Main characteristics of fifth generation

Main electronic component	Based on artificial intelligence, uses the Ultra Large-Scale Integration (ULSI) technology and parallel processing method (ULSI has millions of transistors on a single microchip and Parallel processing method use two or more microprocessors to run tasks simultaneously).
Language	Understand natural language (human language).
Size	Portable and small in size.
Input / output device	Trackpad (or touchpad), touchscreen, pen, speech input (recognize voice/speech), light scanner, printer, keyboard, monitor, mouse, etc.
Example of fifth generation	Desktops, laptops, tablets, smartphones, etc.